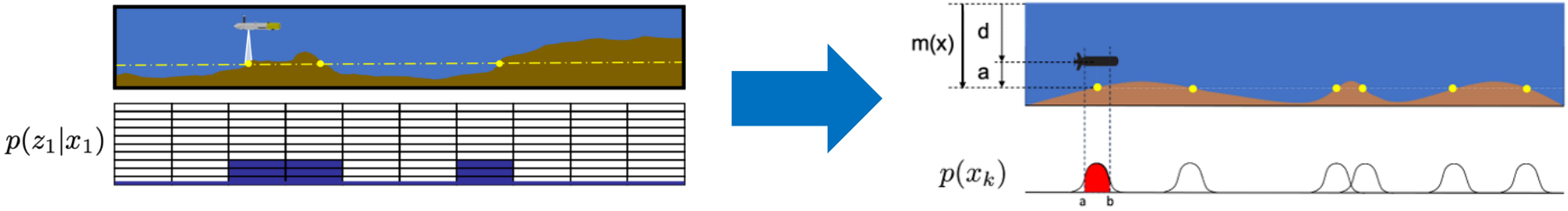
Continuous Probability for Robot Localization

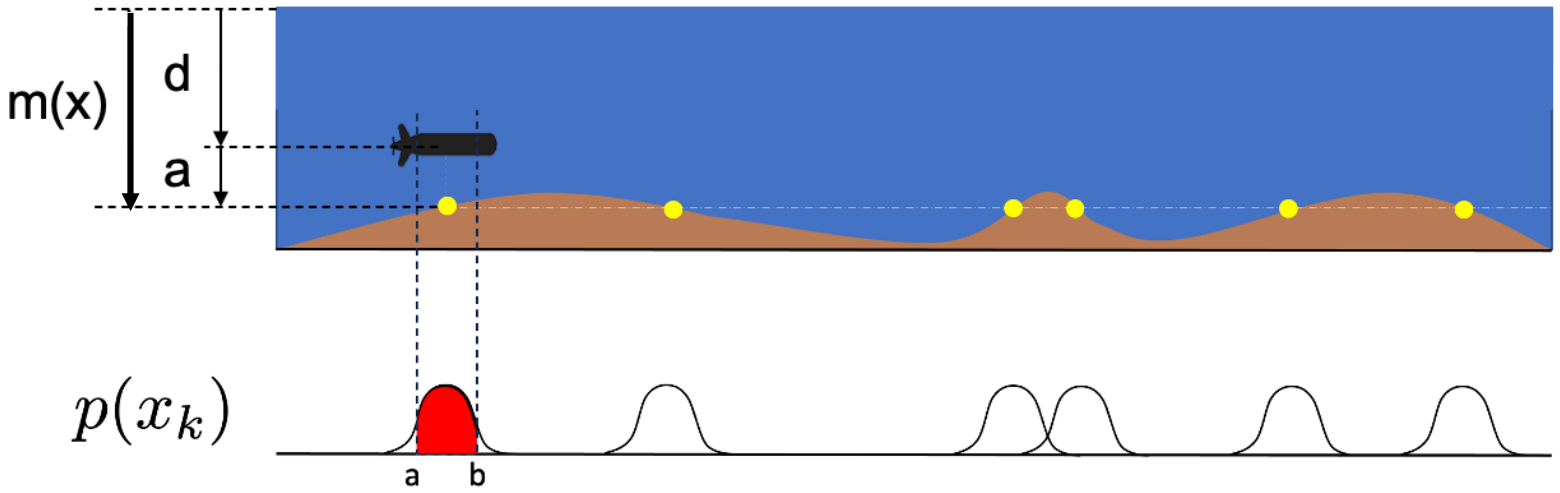
From densities to the continuous Bayes filter, with an AUV over the seafloor



Continuous Probability

Terrain-Based Navigation Example

An AUV localizes itself by measuring the seafloor below it



Measurement model

$$z = d + a = m(x)$$

The vehicle compares the measured terrain depth with a bathymetric map m(x).

- **Ambiguity:** many positions share the same terrain depth, so one measurement gives a multimodal belief.
- **Motion + more measurements** remove the wrong hypotheses until one remains.

Continuous Probability

Lecture Roadmap

Six steps from the die to the seafloor

1

Continuous events

Sample spaces, Borel σ-algebra, random variables

2

Densities and moments

pdf, expected value, variance, uniform, Gaussian, cdf

3

Multivariate densities

Joint pdf, covariance, conditioning, marginals, entropy

4

Inference theorems

Continuous partition, total probability, Bayes' rule

5

Continuous Bayes filter

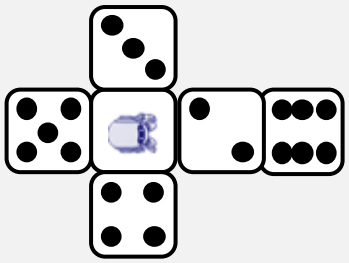
Markov property, Chapman–Kolmogorov, AUV example

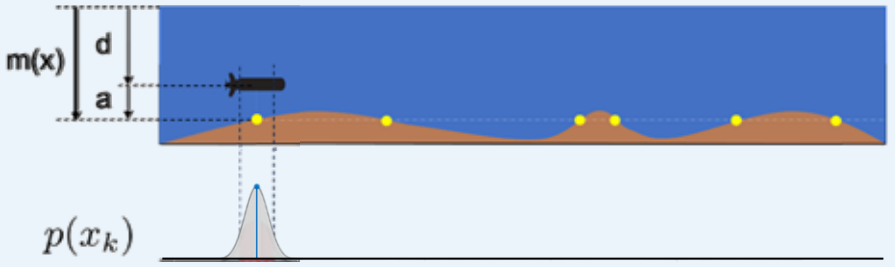
6

Localization & SLAM

Taxonomy of estimation problems and what comes next

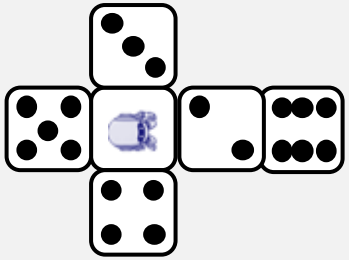


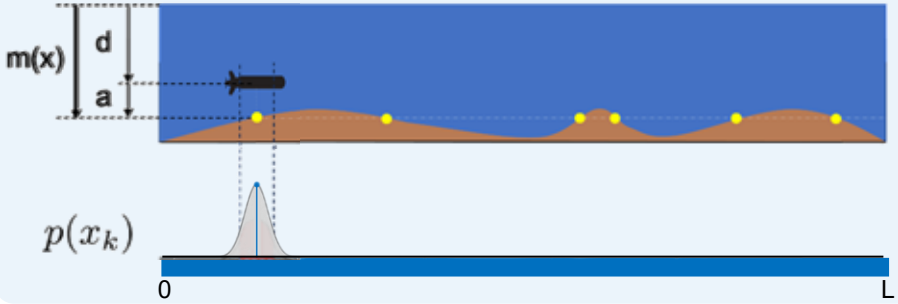




Discrete · robot on a die		Continuous · AUV over the seafloor
$\Omega_X = \{\boxdot, \boxdot, \ldots, \boxdot\}$ (finite)	Sample space	$\Omega = [0, L] \subset \mathbb{R}$ (uncountable)
2^{Ω_X} (every subset)	Events	$\mathcal{B}(\Omega)$ (Borel sets: intervals and combinations)
PMF: $p(x) = P(X = x), \quad \sum_x p(x) = 1$	Distribution	pdf: $p(x) \geq 0, \quad \int p(x)\,dx = 1, \quad P(X = x) = 0$
$P(\mathcal{A}) = \sum_{x \in \mathcal{A}} p(x)$	Prob. of an event	$P(\mathcal{A}) = \int_{\mathcal{A}} p(x)\,dx$
$E[X] = \sum_x x\, p(x)$	Expectation	$E[X] = \int x\, p(x)\,dx$
$p(z) = \sum_x p(z \mid x)\, p(x)$	Total probability	$p(z) = \int p(z \mid x)\, p(x)\,dx$
$p(x \mid z) = \eta\, p(z \mid x)\, p(x), \quad \eta^{-1} = \sum_{x'} p(z \mid x')\, p(x')$	Bayes' rule	$p(x \mid z) = \eta\, p(z \mid x)\, p(x), \quad \eta^{-1} = \int p(z \mid x')\, p(x')\,dx'$
$\overline{\text{bel}}(x_k) = \sum_{x_{k-1}} p(x_k \mid x_{k-1}, u_k)\ \text{bel}(x_{k-1})$	Prediction step	$\overline{\text{bel}}(x_k) = \int p(x_k \mid x_{k-1}, u_k)\ \text{bel}(x_{k-1})\,dx_{k-1}$





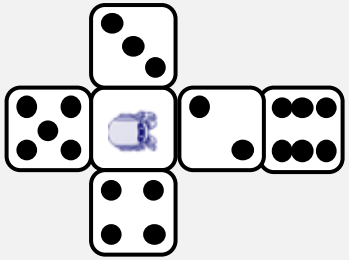


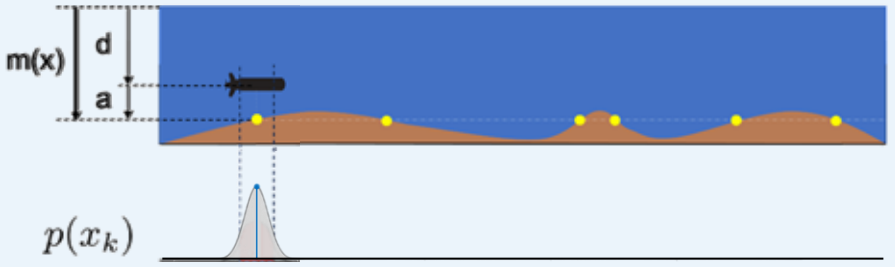
$\Omega_X = \{\boxdot, \boxdot, \ldots, \boxdot\}$ (finite)

Sample space

$\Omega = [0, L] \subset \mathbb{R}$ (uncountable)





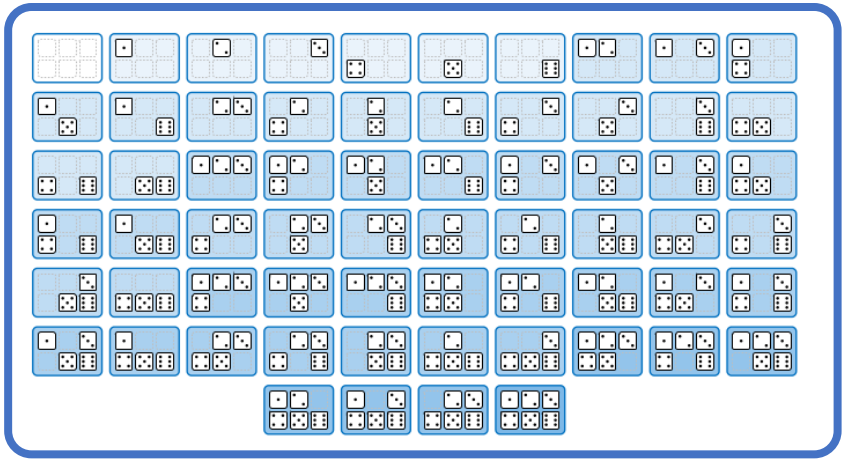


2^{Ω_X} (every subset)

Events

$\mathcal{B}(\Omega)$ (Borel sets: intervals and combinations)

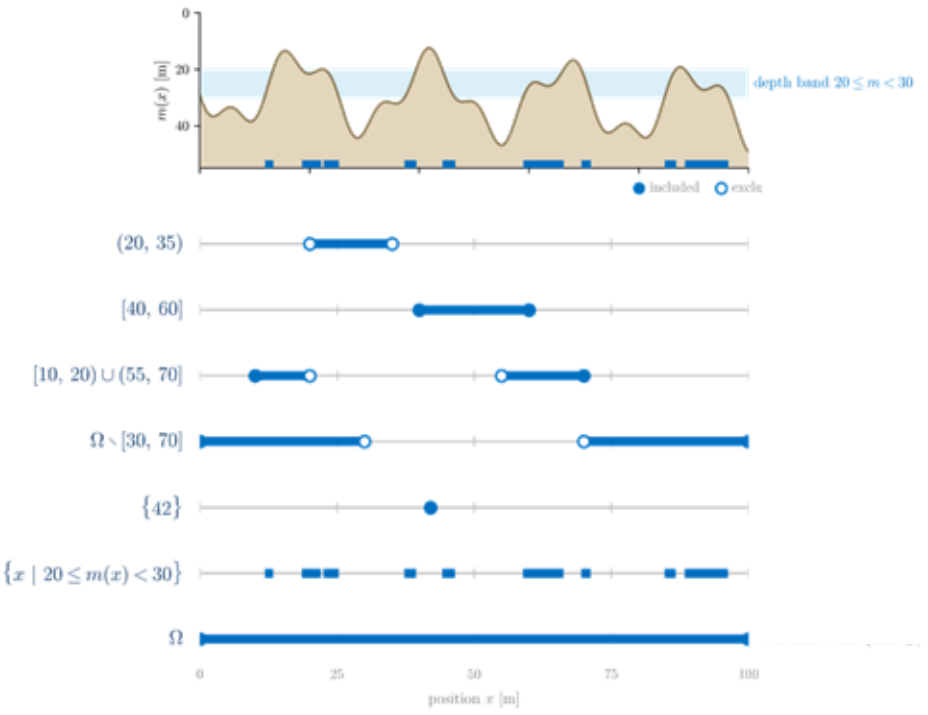
$$\Sigma_\Omega = 2^\Omega$$



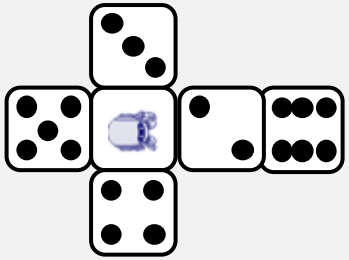
$\mathcal{B}(\Omega) = \sigma(\{\text{open intervals of } \Omega\})$ $\bigcup_i \mathcal{A}_i \in \mathcal{B}$, $\bigcap_i \mathcal{A}_i \in \mathcal{B}$, $\mathcal{A}^c \in \mathcal{B}$

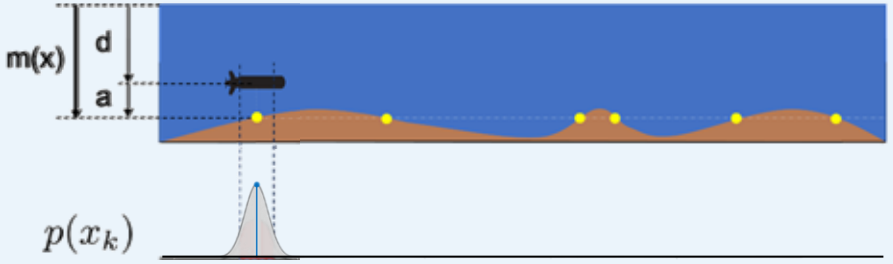
$\mathcal{B}(\Omega)$: the Borel sets of the AUV transect $\Omega = [0, L]$, $L = 100$ m

every region of the seafloor we may ask a probability about





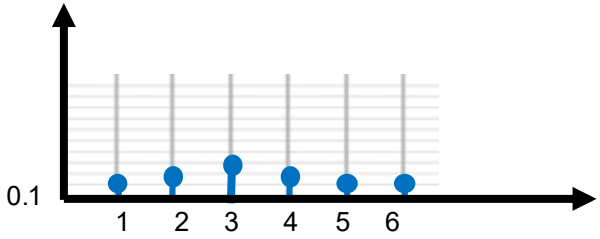




PMF: $p(x) = P(X = x), \quad \sum_x p(x) = 1$

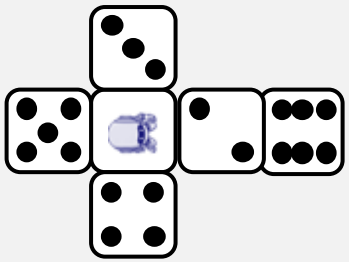
Distribution

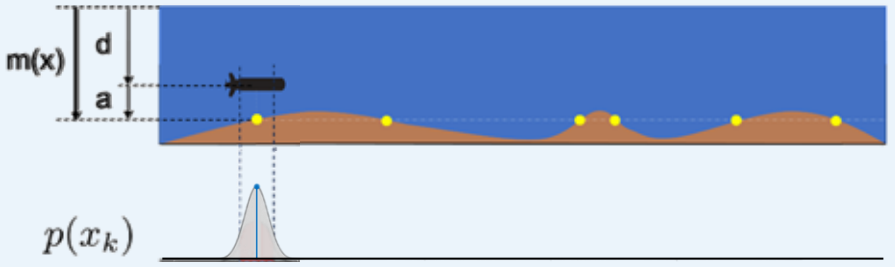
pdf: $p(x) \geq 0, \quad \int p(x)\,dx = 1, \quad P(X = x) = 0$







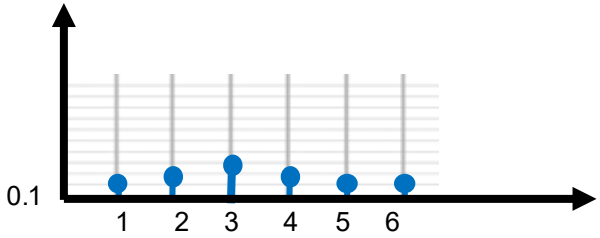




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Distribution

pdf: $p(x) \geq 0, \quad \int p(x)\,dx = 1, \quad P(X = x) = 0$





$$P(\mathcal{A}) = \sum_{x \in \mathcal{A}} p(x)$$

Prob. of an event

$$P(\mathcal{A}) = \int_{\mathcal{A}} p(x)\,dx$$

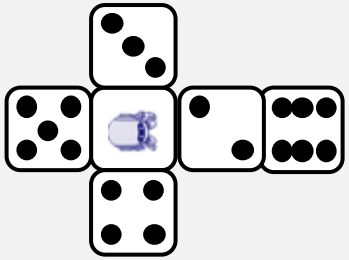
Even face: $\mathcal{A} = \{\boxdot, \boxdot, \boxdot\}$

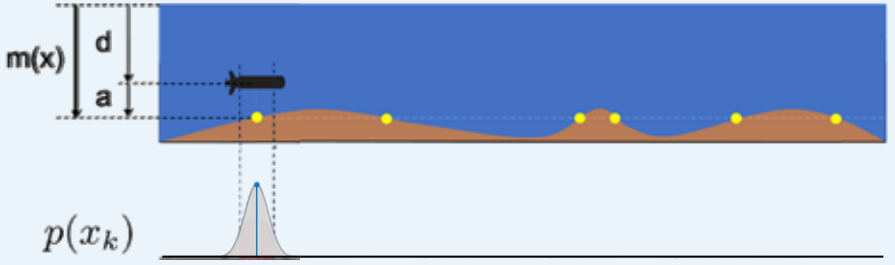
$$\begin{aligned} P(\mathcal{A}) &= P(\{\boxdot\}) + P(\{\boxdot\}) + P(\{\boxdot\}) \\ &= 0.2 + 0.2 + 0.1 \\ &= \frac{3}{6} = 0.5 \end{aligned}$$

1st half: $\mathcal{A} = [a, b]$

$$P(\mathcal{A}) = \int_a^b p(x)dx \text{ = area}$$



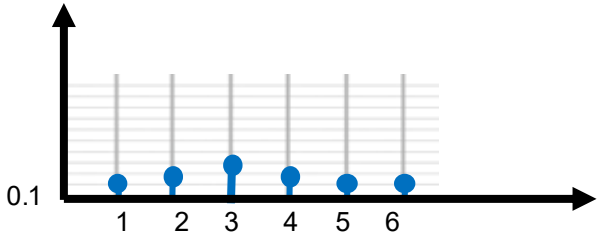




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Distribution

pdf: $p(x) \geq 0, \quad \int p(x)\,dx = 1, \quad P(X = x) = 0$





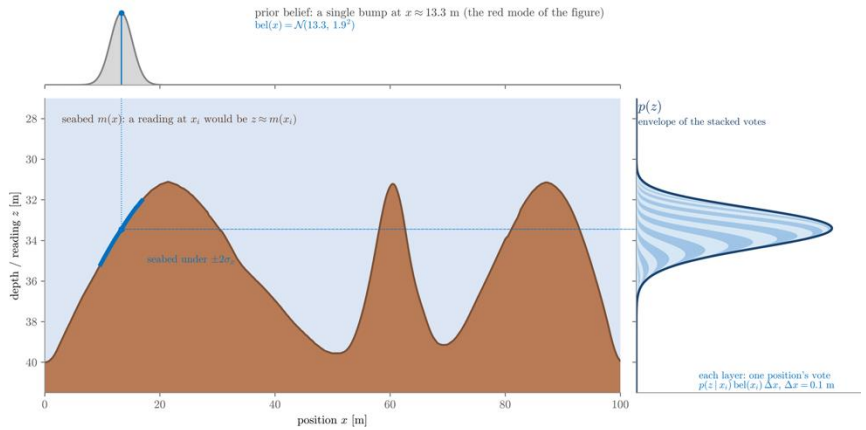
$$p(z) = \sum_x p(z \mid x)\, p(x)$$

Total probability

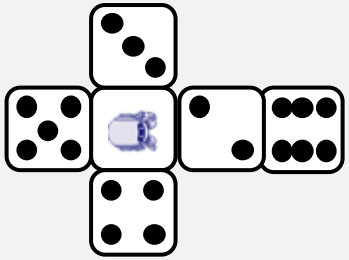
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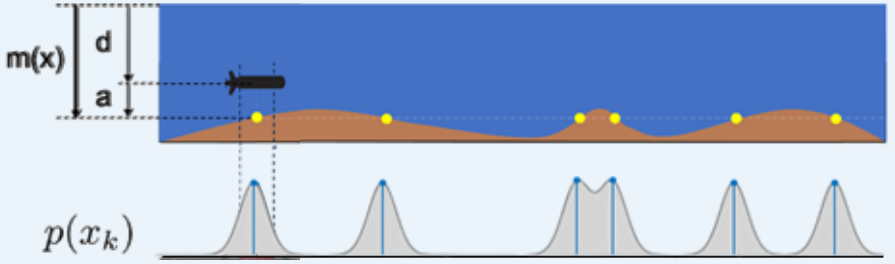
Even face: $\mathcal{A} = \{\boxdot, \boxdot, \boxdot\}$

$$\begin{aligned} P(z = odd) &= P(z = odd|x = 1) + P(z = odd|x = 2) + P(z = odd|x = 3) \\ &\quad + P(z = odd|x = 4) + P(z = odd|x = 5) + P(z = odd|x = 6) \\ &= 1 \cdot 0.1 + 0 \cdot 0.2 + 1 \cdot 0.3 + 0 \cdot 0.2 + 1 \cdot 0.1 + 0 \cdot 0.1 \\ &= 0.5 \end{aligned}$$





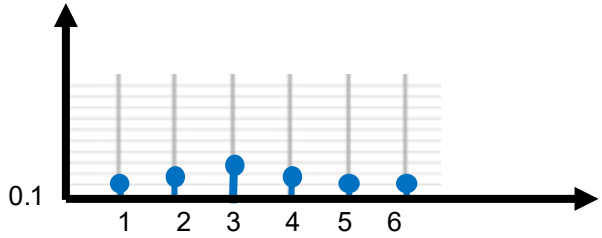




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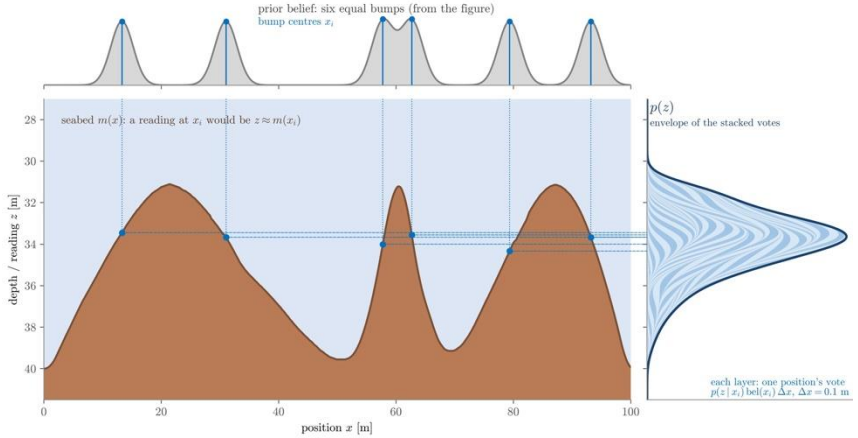
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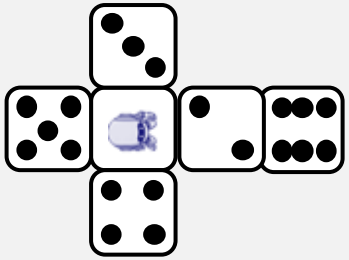
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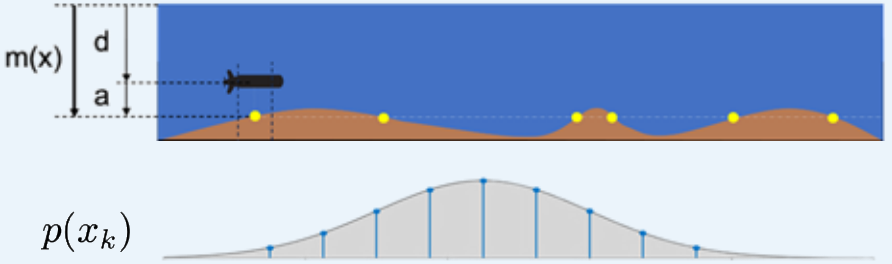
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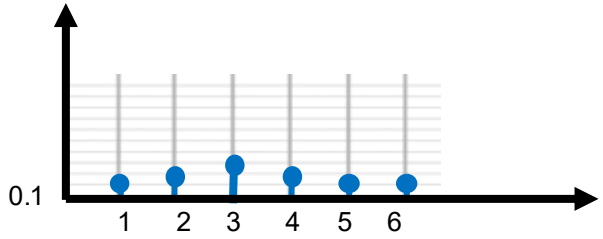


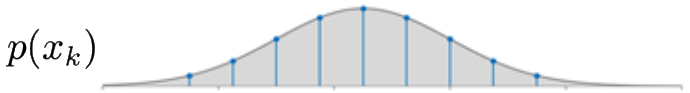


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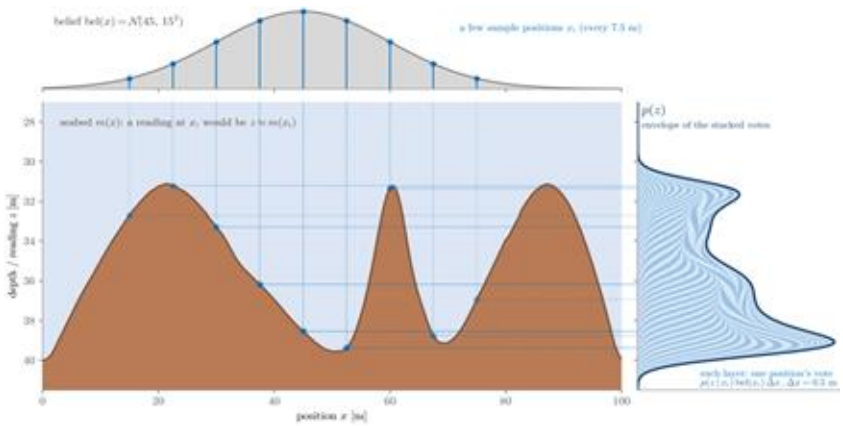
$$p(z) = \sum_x p(z \mid x)\, p(x)$$

Total probability

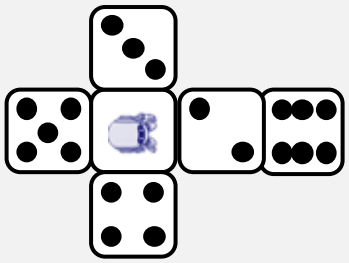
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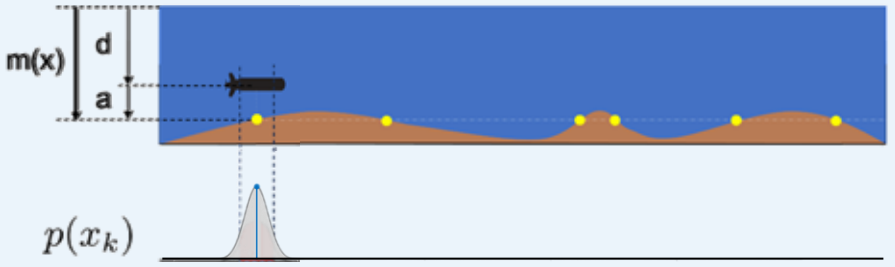
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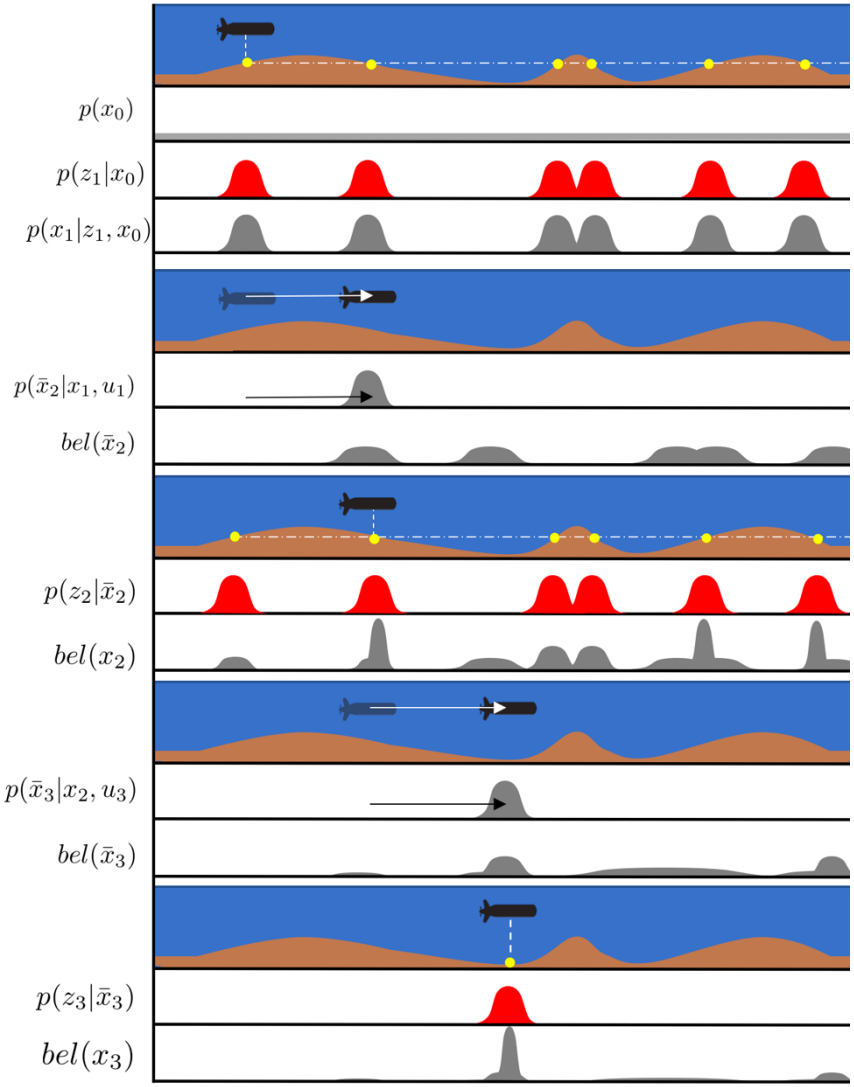


PMF: $p(x) = P(X = x), \quad \sum_x p(x) = 1$	**Distribution**	pdf: $p(x) \geq 0, \quad \int p(x)\,dx = 1, \quad P(X = x) = 0$
$p(z) = \sum_x p(z \mid x)\, p(x)$	**Total probability**	$p(z) = \int p(z \mid x)\, p(x)\, dx$
$\overline{\text{bel}}(x_k) = \sum_{x_{k-1}} p(x_k \mid x_{k-1}, u_k)\ \text{bel}(x_{k-1})$	**Prediction step**	$\overline{\text{bel}}(x_k) = \int p(x_k \mid x_{k-1}, u_k)\ \text{bel}(x_{k-1})\, dx_{k-1}$
$p(x \mid z) = \eta\, p(z \mid x)\, p(x), \quad \eta^{-1} = \sum_{x'} p(z \mid x')\, p(x')$	**Bayes' rule**	$p(x \mid z) = \eta\, p(z \mid x)\, p(x), \quad \eta^{-1} = \int p(z \mid x')\, p(x')\, dx'$

Continuous Probability

Markov Localization over the Seafloor

Prediction spreads the belief; each measurement prunes hypotheses



- Start: uniform, no idea where the AUV is.
- First measurement: every position with that terrain depth becomes a peak.
- Moving 12 m shifts the peaks and widens them.
- Each new measurement keeps only consistent peaks.
- After three measurements one hypothesis survives.

Continuous Probability

Continuous Random Variables

A random variable turns the AUV's physical position into a number

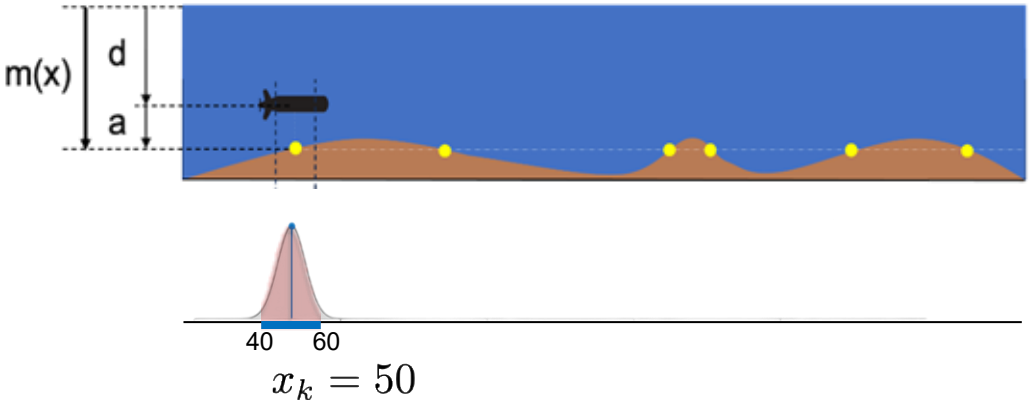
$$X : \Omega \to \mathbb{R}, \qquad \{\omega \in \Omega \mid X(\omega) \leq x\} \in \mathcal{B}(\Omega)$$

Bridge between world and map

If the AUV is physically at ω, X(ω) is the coordinate (e.g. 15.42 m) the filter processes.

Consequence

We cannot assign probability to a single value X = x. We need a density, and probabilities come from integrating it over intervals.



$$P(x_k = 50) = \int_0^0 p(x_k)dx_k = 0$$

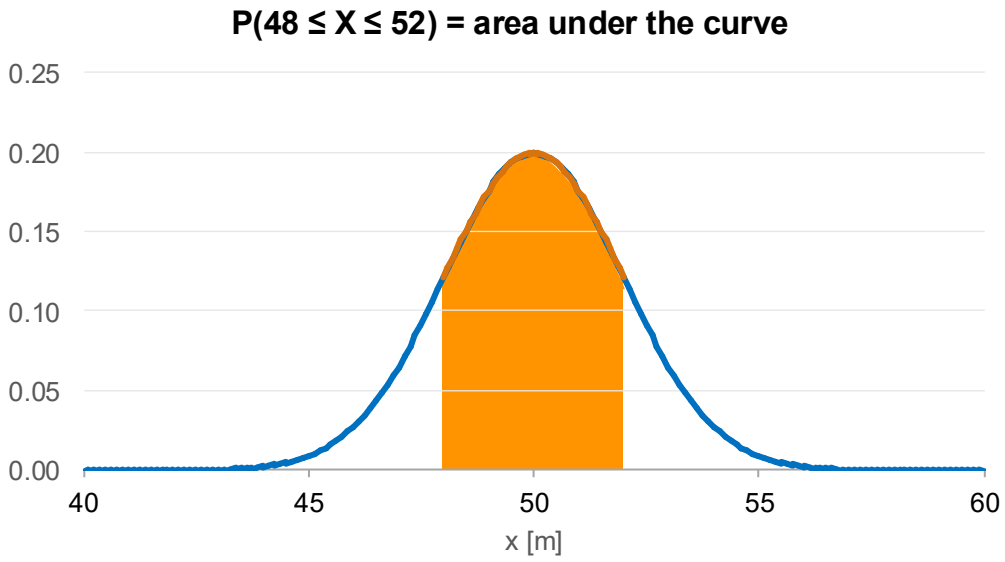
$$P(40 \leq x_k \leq 60) = \int_{40}^{60} p(x_k)dx_k \neq 0$$

Continuous Probability

Probability Density Function

Probability is the area under the density

$$P(\mathcal{A}) = \int_{\mathcal{A}} p(x)\,dx, \qquad p(x) \geq 0, \qquad \int_{-\infty}^{\infty} p(x)\,dx = 1$$



Read with care

- p(x) is a density: probability per metre.
- It can be larger than 1 for a very precise belief.
- Only integrals of p(x) are probabilities.

Moments: the Expected Value

The mean is the robot's best guess of where it is

$$E[X] = \mu = \int_{-\infty}^{\infty} x\, p(x)\, dx$$

Properties

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$
$$E[c] = c$$
$$X \leq Y \Rightarrow E[X] \leq E[Y]$$
$$X \perp Y \Rightarrow E[XY] = E[X]E[Y]$$

Example: metres to feet

$$E[3.28X] = 3.28\, E[X]$$

With Y = 3.28 X, linearity scales the mean directly: no new integral is needed.

Continuous Probability

Moments: Variance and Standard Deviation

The variance measures how uncertain the robot is

$$\mathrm{Var}(X) = \sigma^2 = E\left[(X-\mu)^2\right] = \int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx$$

Properties

$$\mathrm{Var}(X) \geq 0, \qquad \mathrm{Var}(a) = 0$$
$$\mathrm{Var}(X+a) = \mathrm{Var}(X)$$
$$\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$$
$$X \perp Y \Rightarrow \mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$
$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Example: metres to feet

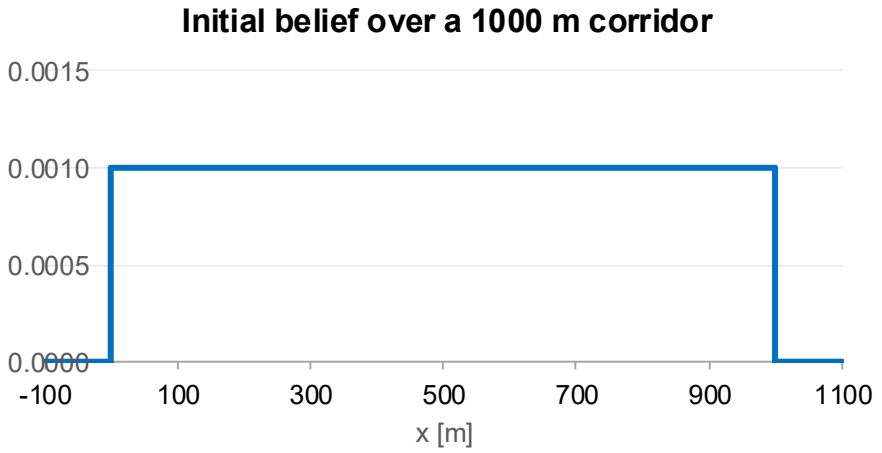
$$\mathrm{Var}(3.28X) = 3.28^2\,\mathrm{Var}(X) \approx 10.76\,\mathrm{Var}(X)$$

The mean scales by 3.28, the variance by 3.28². Adding a constant offset (sensor to keel) leaves the variance unchanged.

The Uniform Distribution

A uniform density models “somewhere in here, no idea where”

$$p(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$



Global uncertainty, L = 1000 m

$$\text{bel}(x_0) = \frac{1}{1000} = 0.001 \text{ m}^{-1}, \qquad P(0 \le X \le 250) = \int_0^{250} 0.001\, dx = 0.25$$

A 25 % chance of being in the first quarter, as the area of the rectangle says.

Moments of the Uniform Distribution

Best guess: the midpoint. Uncertainty: almost a third of the corridor

Mean $$E[X] = \int_0^{1000} \frac{x}{1000}\,dx = \frac{1000}{2} = 500 \text{ m}$$

Variance $$\mathrm{Var}(X) = \underbrace{\frac{1000^2}{3}}_{E[X^2]} - \underbrace{500^2}_{(E[X])^2} = \frac{1000^2}{12} \approx 83\,333 \text{ m}^2$$

500 m

Expected position: the corridor's midpoint

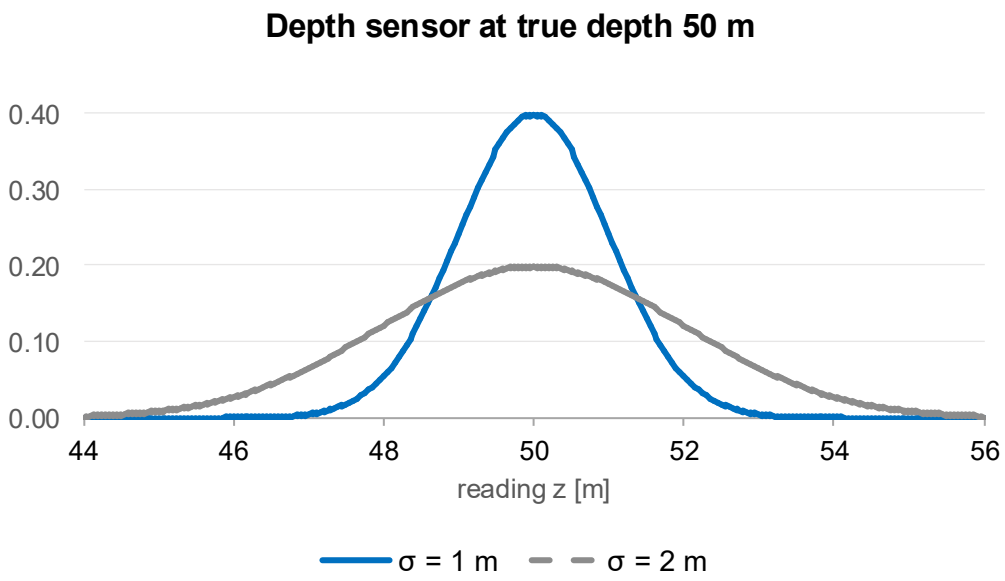
288.7 m

Standard deviation: σ = 1000/√12

The Gaussian (Normal) Distribution

The Gaussian is the workhorse model of sensor noise

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad X \sim \mathcal{N}(\mu, \sigma^2)$$



Reading the curve

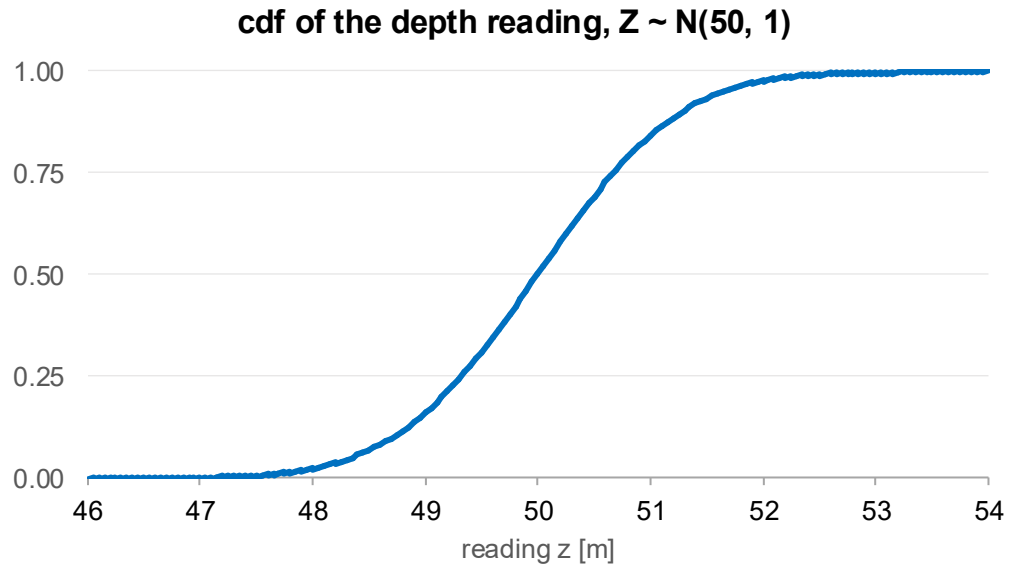
$Z \sim \mathcal{N}(50, 1^2)$

- Peak at 50 m: the most likely reading.
- 50.5 m is very plausible; 60 m is 10σ away.
- Better sensor: narrower and taller curve.

Cumulative Distribution Function

The cdf accumulates probability up to x

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} p(t)\,dt, \qquad P(a \leq X \leq b) = F(b) - F(a)$$



Within ±1σ

$$P(49 \leq Z \leq 51) = F(51) - F(49) \approx 0.841 - 0.159 = 0.683$$

About 68 % of the readings fall within one standard deviation of the true depth.

The Continuous Probability Space

Same triple as for the die, with the Borel σ-algebra

$$(\Omega,\ \mathcal{B}(\Omega),\ P)$$

Ω

the continuous sample space: the AUV's coordinates

$\mathcal{B}(\Omega)$

the Borel σ-algebra: all measurable regions

P

the measure: integrates the pdf over a region

Why Gaussians dominate robotics

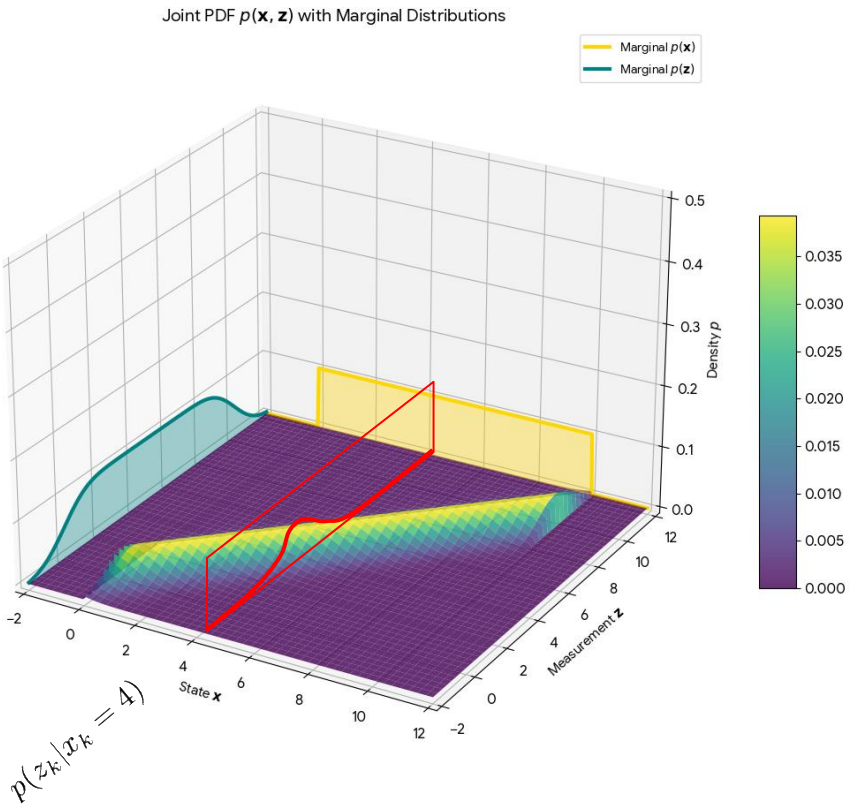
- Closure: sums and linear maps of Gaussians stay Gaussian, and two numbers (μ, σ^2) describe them.
- Central Limit Theorem: sums of many small independent errors tend to a Gaussian.

Continuous Probability

Joint Probability Density Functions

A joint density couples the state and the measurement

$$P\big((X,Z)\in\mathcal{A}\big)=\iint_{\mathcal{A}} p(x,z)\,dx\,dz, \qquad \iint p(x,z)\,dx\,dz=1$$



AUV depth and pressure sensor

$$p(x)=\mathcal{U}(0,10)$$

$$p(z\mid x)=\mathcal{N}(z;\,x,\,1)$$

$$p(x,z)=p(z\mid x)\,p(x)=$$

$$=\frac{1}{10\sqrt{2\pi}}\,e^{-\frac{(z-x)^2}{2}}, \quad 0\le x\le 10$$

A diagonal ridge, clipped at the prior's edges.

Covariance and correlation quantify the coupling

$$\boldsymbol{\mu} = \begin{bmatrix} E[x] \\ E[z] \end{bmatrix}, \qquad \sigma_{xz} = E\big[(x-\mu_x)(z-\mu_z)\big], \qquad \rho = \frac{\sigma_{xz}}{\sigma_x \sigma_z} \in [-1, 1]$$

AUV depth and sensor reading (z = x + v)

$$\sigma_x^2 = \frac{10^2}{12} \approx 8.33, \quad \sigma_z^2 = \sigma_x^2 + \sigma_v^2 \approx 9.33, \quad \sigma_{xz} = \sigma_x^2 \approx 8.33$$

$$\rho = \frac{8.33}{\sqrt{8.33}\,\sqrt{9.33}} \approx 0.94$$

Near-perfect correlation: z is a very good proxy for the hidden x.

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_x^2 & \sigma_{xz} \\ \sigma_{zx} & \sigma_z^2 \end{bmatrix} = \begin{bmatrix} 8.33 & 8.33 \\ 8.33 & 9.33 \end{bmatrix}$$

Conditional Density

Conditioning slices the joint density and renormalizes

$$p(x \mid z) = \frac{p(x, z)}{p(z)}, \qquad p(z) > 0$$

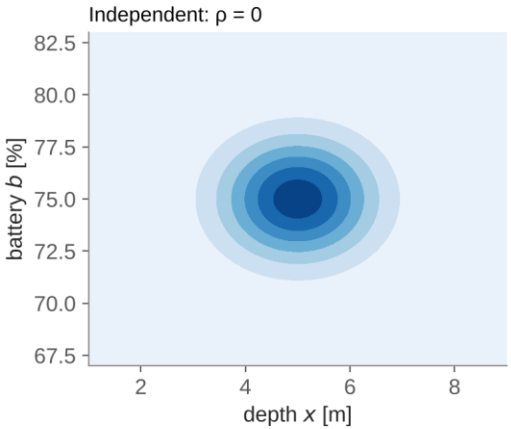
Slicing the AUV joint density at a fixed x

$$p(z \mid x) = \frac{p(x, z)}{p(x)} = \frac{\frac{1}{10}\,\frac{1}{\sqrt{2\pi}} e^{-\frac{(z-x)^2}{2}}}{\frac{1}{10}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-x)^2}{2}}$$

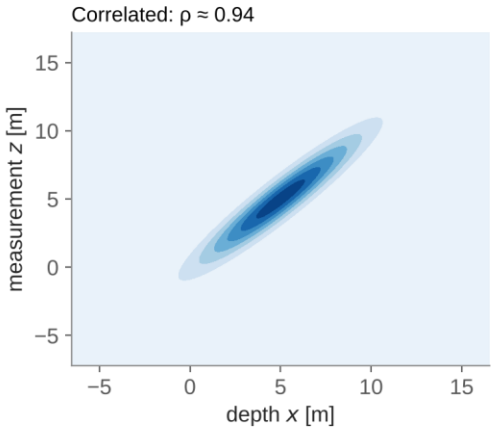
- **This is the engine of localization:** after a measurement z the robot wants p(x | z).
- Here the slice p(z | x) is just the sensor's noise model, independent of the prior.

Independent variables: the joint density factorizes

$$X \perp Y \iff p(x,y) = p(x)\,p(y) \iff p(x \mid y) = p(x)$$



independent, axis-aligned.



the correlated AUV depth/sensor pair.

Depth x and battery b

Uniform priors: a flat plateau.

$$p(x,b) = \frac{1}{10} \cdot \frac{1}{100} = \frac{1}{1000},$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 8.33 & 0 \\ 0 & 833.33 \end{bmatrix}$$

Gaussian estimates: an axis-aligned mound.

$$x \sim \mathcal{N}(5, 1^2),\ b \sim \mathcal{N}(75, 2^2),$$

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$$

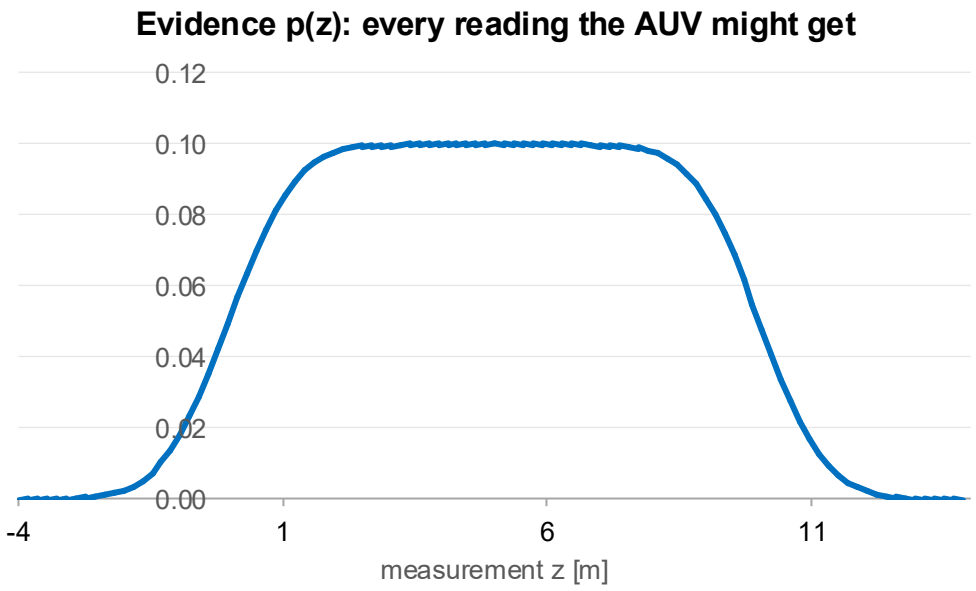
Zero off-diagonal terms: ρ = 0.

Continuous Probability

Marginalization in Continuous Spaces

Integrating out a variable projects the joint onto one axis

$$p(x) = \int_{-\infty}^{\infty} p(x, z)\, dz, \qquad p(z) = \int_{-\infty}^{\infty} p(x, z)\, dx$$



Recover the state prior

$$p(x) = p(x) \underbrace{\int p(z \mid x)\, dz}_{=1}$$

Evidence of the reading

$$p(z) = \int_0^{10} \frac{1}{10\sqrt{2\pi}}\, e^{-\frac{(z-x)^2}{2}}\, dx$$

A broadened Gaussian: location uncertainty + sensor noise.

Differential Entropy

Differential entropy measures the spread of a continuous belief

$$h(\boldsymbol{x}) = -\int p(\boldsymbol{x}) \ln p(\boldsymbol{x})\, d\boldsymbol{x}$$

Gaussian belief

$$h(\boldsymbol{x}) = \tfrac{1}{2} \ln\big((2\pi e)^n \det \boldsymbol{\Sigma}\big) = \tfrac{n}{2} \ln(2\pi e) + \tfrac{1}{2} \ln \det \boldsymbol{\Sigma}$$

$$\det \boldsymbol{\Sigma} = \prod_{i=1}^{n} \lambda_i, \qquad \ln \det \boldsymbol{\Sigma} = \sum_{i=1}^{n} \ln \lambda_i$$

3.72 nats

Prior N(20, 10^2)

3.74 nats

After prediction N(30, 104)

2.92 nats

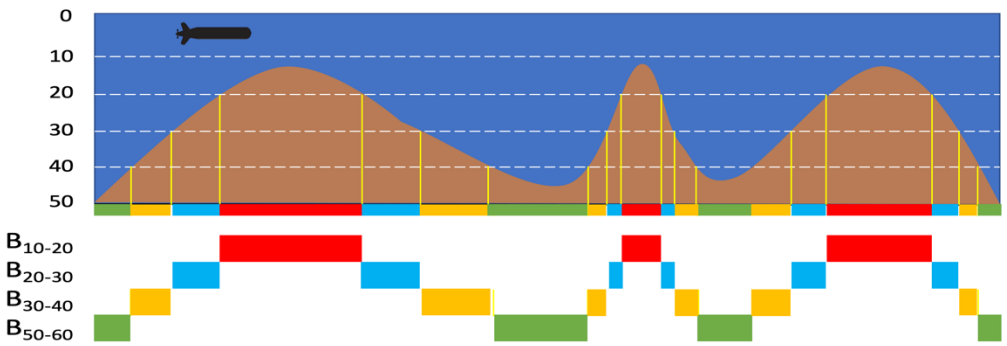
After the update N(34.0, 20.2)

Unlike discrete entropy, it can be negative and depends on units: compare beliefs in the same coordinates.

Continuous Partition of the Sample Space

Partition the seafloor by map depth

$$\mathcal{B}_i \neq \emptyset, \qquad \mathcal{B}_i \cap \mathcal{B}_j = \emptyset \ (i \neq j), \qquad \bigcup_i \mathcal{B}_i = \Omega$$



Sum over bands → integral over every x.

One band, many intervals

$$\mathcal{B}_{10-20} = \{x \in \Omega \mid 10 \leq m(x) < 20\}$$

Shrinking the bands:

$$\mathcal{B}_d = \{x \in \Omega \mid d \leq m(x) < d + \Delta d\}, \quad \Delta d \to 0$$

Each coordinate x becomes its own element of the partition.

Continuous Probability

The Continuous Law of Total Probability

The evidence of a measurement integrates over all states

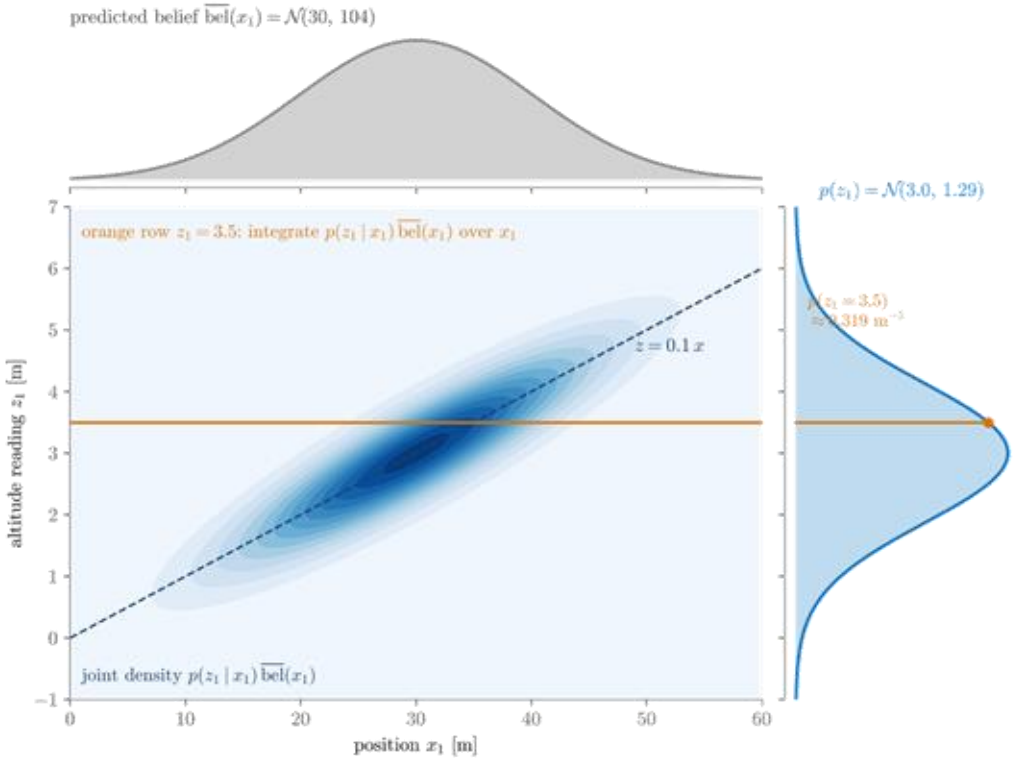
Discrete partition

$$P(\mathcal{A}) = \sum_i P(\mathcal{A} \mid \mathcal{B}_i)\, P(\mathcal{B}_i)$$

Continuous state

$$p(z) = \int_\Omega p(z \mid x)\, p(x)\, dx$$

- Product rule: p(x, z) = p(z | x) p(x), the joint density of slide 19.
- Marginalizing x gives p(z): the curve on the marginalization slide.
- We usually start from a prior and a sensor model, not from a joint density.



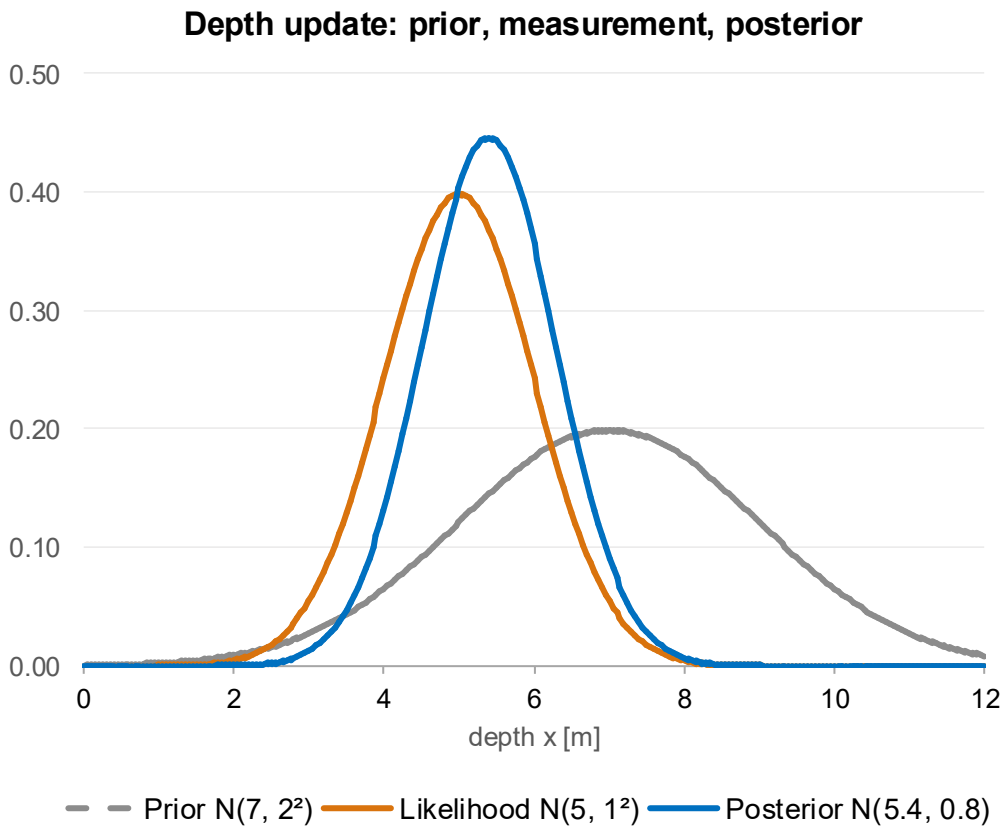
Bayes' Rule for Continuous Variables

Bayes' rule turns a measurement into a posterior density

$$\underbrace{p(x \mid z)}_{\text{posterior}} = \frac{\overbrace{p(z \mid x)}^{\text{likelihood}}\ \overbrace{p(x)}^{\text{prior}}}{\underbrace{p(z)}_{\text{evidence}}} = \eta\, p(z \mid x)\, p(x) \qquad \eta = \left[\int_{\Omega} p(z \mid x')\, p(x')\, dx'\right]^{-1}$$

- Prior p(x): belief before the measurement.
- Likelihood p(z | x): sensor model and map m(x).
- Evidence p(z): from the Law of Total Probability; it only normalizes.
- On the seafloor, multiplying by the prior removes peaks at inconsistent positions.

Fusing two Gaussians: a precision-weighted average



Closed form

$$\sigma^2_{\text{post}} = \frac{\sigma_x^2 \sigma_z^2}{\sigma_x^2 + \sigma_z^2}, \qquad \mu_{\text{post}} = \sigma^2_{\text{post}} \left(\frac{\mu_x}{\sigma_x^2} + \frac{z}{\sigma_z^2} \right)$$

$$\sigma^2_{\text{post}} = \frac{4 \cdot 1}{4 + 1} = 0.8, \qquad \mu_{\text{post}} = 0.8 \left(\frac{7}{4} + \frac{5}{1} \right) = 5.4$$

- **Mean shift:** 5.4 is pulled toward the more precise sensor.
- **Less uncertainty:** 0.8 < 1 < 4.

The Markov property keeps the belief at a fixed size

$$p\big(x(t) \mid x(\tau),\ \tau \le s\big) = p\big(x(t) \mid x(s)\big), \qquad t > s$$

Non-Markovian: the trajectory

$$\overline{\text{bel}}(x_{0:2}) = p(x_2 \mid u_{1:2}, x_0, x_1)\ \text{bel}(x_0, x_1)$$

$$\boldsymbol{\Sigma}_2 = \begin{bmatrix} \sigma^2_{x_0} & \sigma_{x_0 x_1} & \sigma_{x_0 x_2} \\ \sigma_{x_1 x_0} & \sigma^2_{x_1} & \sigma_{x_1 x_2} \\ \sigma_{x_2 x_0} & \sigma_{x_2 x_1} & \sigma^2_{x_2} \end{bmatrix}$$

The belief gains a dimension every step: cost grows with t.

Markovian: only the present

$$\overline{\text{bel}}(x_2) = \int p(x_2 \mid u_2, x_1)\ \text{bel}(x_1)\, dx_1$$

x_1 is integrated out and discarded. The belief stays one-dimensional: constant cost per step.

The Continuous Recursive Bayes Filter

Same prediction–correction cycle, with integrals

1 Prediction (Chapman–Kolmogorov)

$$\overline{\mathrm{bel}}(x_k) = \int_{\mathcal{X}} p(x_k \mid x_{k-1}, u_k)\ \mathrm{bel}(x_{k-1})\, dx_{k-1}$$

2 Correction (Bayes' rule)

$$\mathrm{bel}(x_k) = \eta_k\, p(z_k \mid x_k)\, \overline{\mathrm{bel}}(x_k), \qquad \eta_k = \left[\int_{\mathcal{X}} p(z_k \mid x')\, \overline{\mathrm{bel}}(x')\, dx' \right]^{-1}$$

The integrals rarely have a closed form: that is why the rest of the course studies histogram, particle and Gaussian filters.

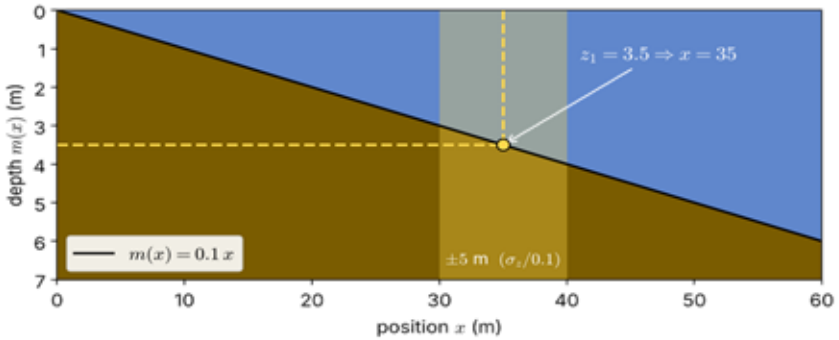
Continuous Probability

Localization in Action: AUV over the Seafloor

A worked iteration: models and prediction

Initial belief

$$p(x_0) = \mathcal{N}(x;\, 20,\, 10^2)$$



Motion: 10 m forward

$$p(x_k \mid x_{k-1}, u_k) = \mathcal{N}(x_k;\, x_{k-1} + 10,\, 2^2)$$

Altitude: map m(x) = 0.1x

$$p(z_k \mid x_k) = \mathcal{N}(z_k;\, 0.1\,x_k,\, 0.5^2)$$

Prediction: means and variances add

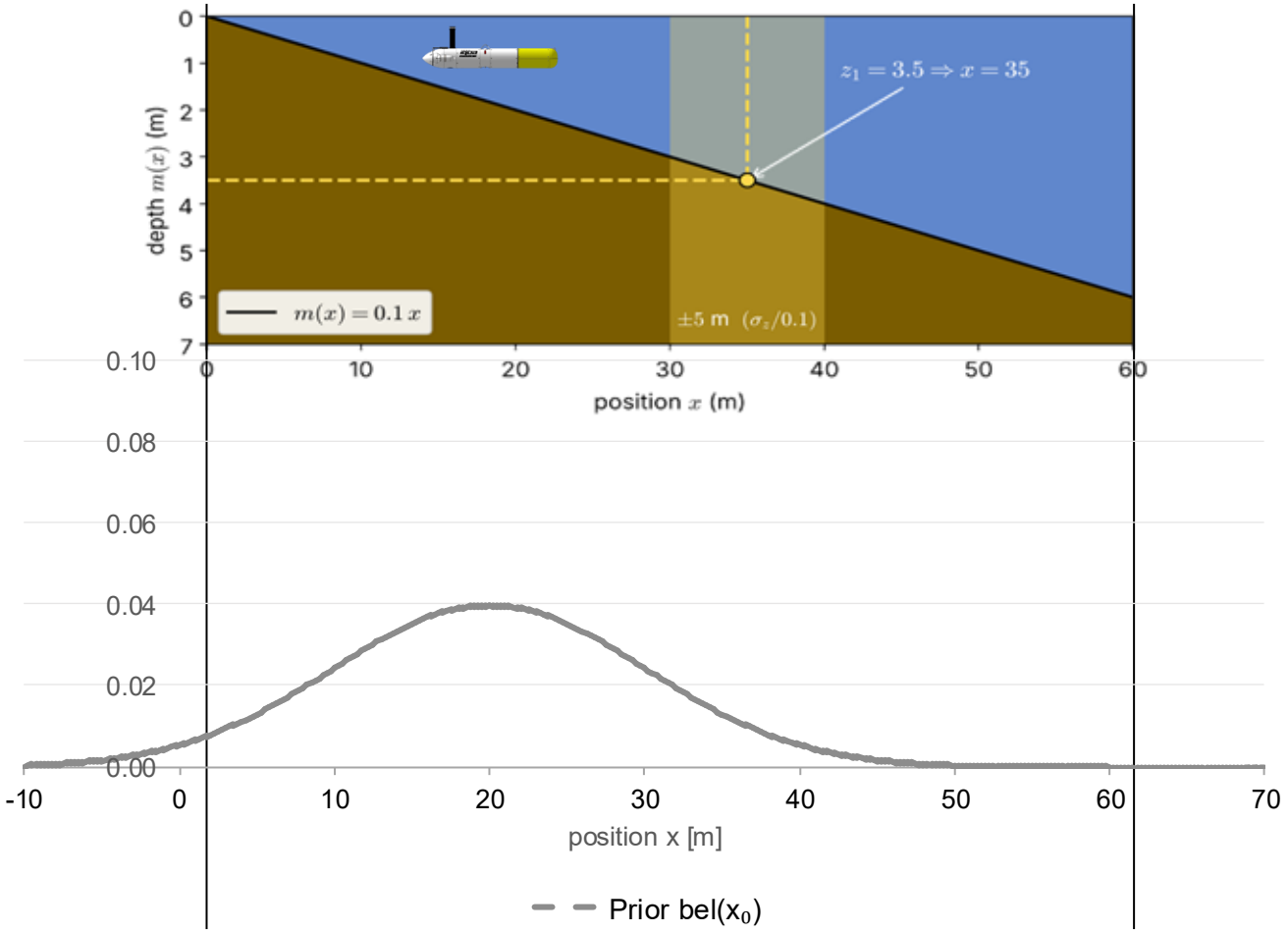
$$\overline{\text{bel}}(x_1) = \mathcal{N}(x;\, 20 + 10,\; 10^2 + 2^2) = \mathcal{N}(x;\, 30,\, 104)$$

Update with z_1 = 3.5 (map depth at x = 35)

$$z_1 = 3.5 \;\Rightarrow\; p(z_1 \mid x) \propto \mathcal{N}(x;\, 35,\, 5^2) \quad \text{bel}(x_1) = \mathcal{N}(x;\, 34.0,\, 20.2)$$

Continuous Probability

AUV Example: One Filter Iteration

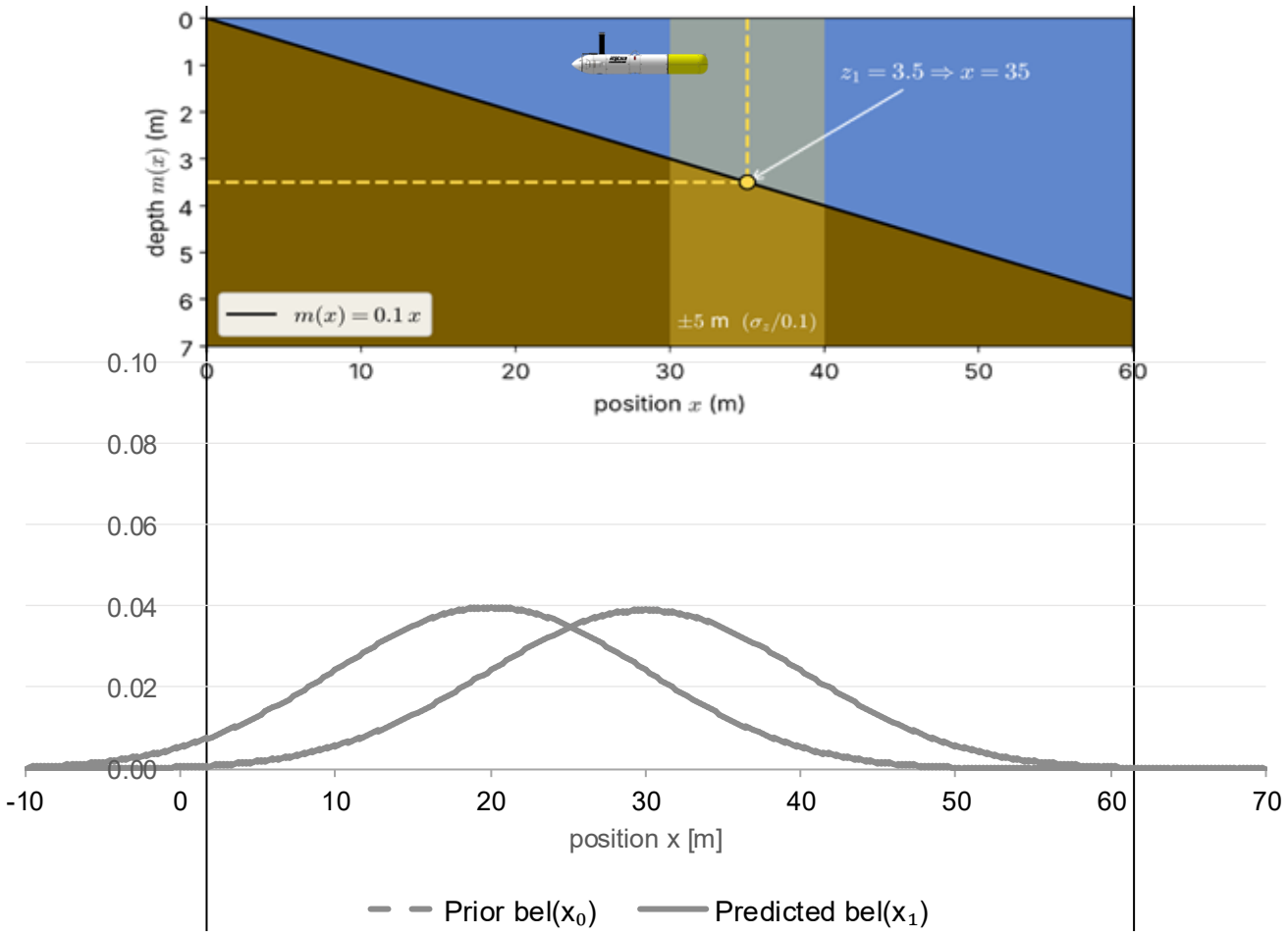


Initial belief

$$p(x_0) = \mathcal{N}(x;\, 20,\, 10^2)$$

Continuous Probability

AUV Example: One Filter Iteration

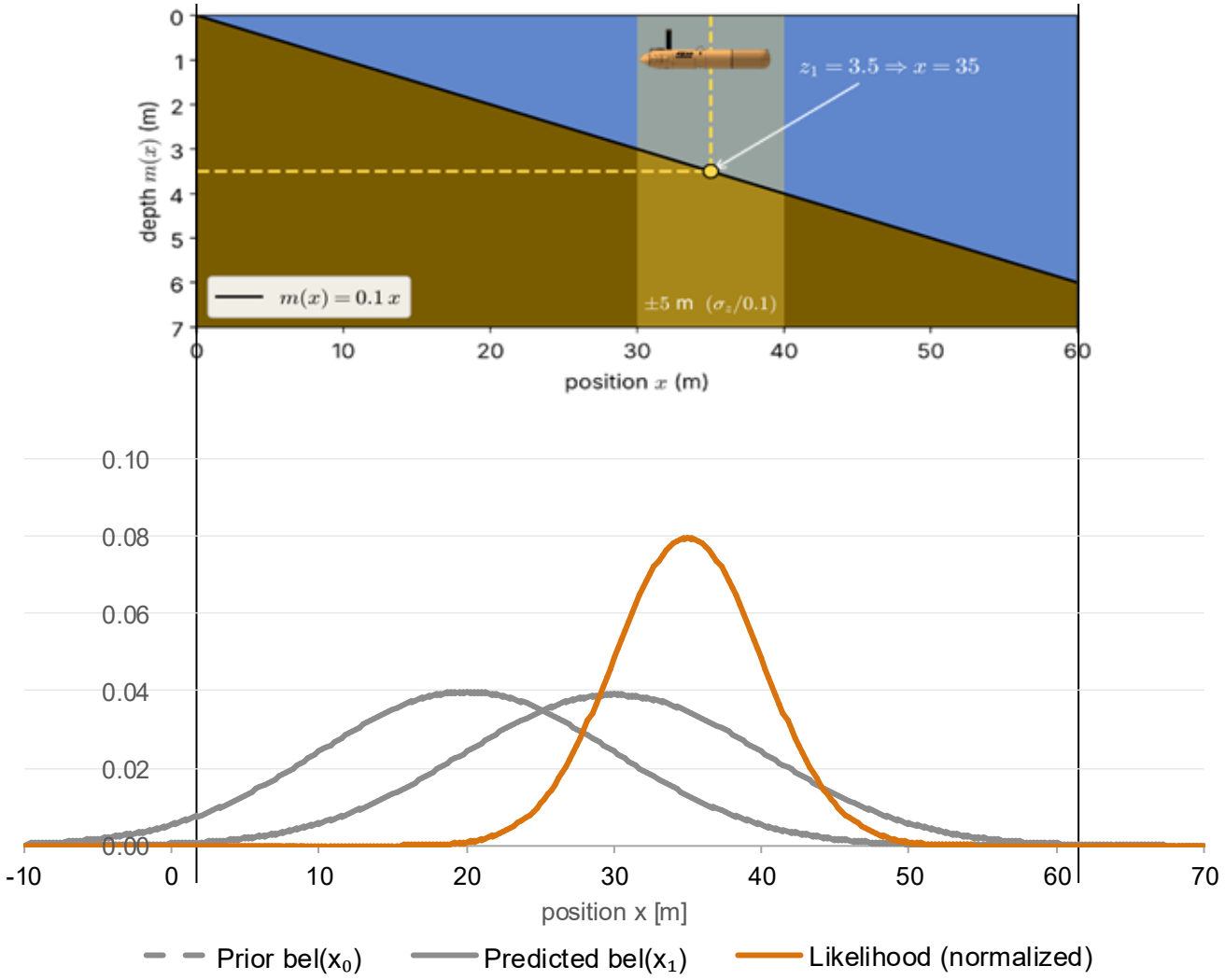


Motion: 10 m forward $p(x_k \mid x_{k-1}, u_k) = \mathcal{N}(x_k;\, x_{k-1} + 10,\, 2^2)$

Prediction: $\overline{\text{bel}}(x_1) = \mathcal{N}(x;\, 20 + 10,\; 10^2 + 2^2) = \mathcal{N}(x;\, 30,\, 104)$

Continuous Probability

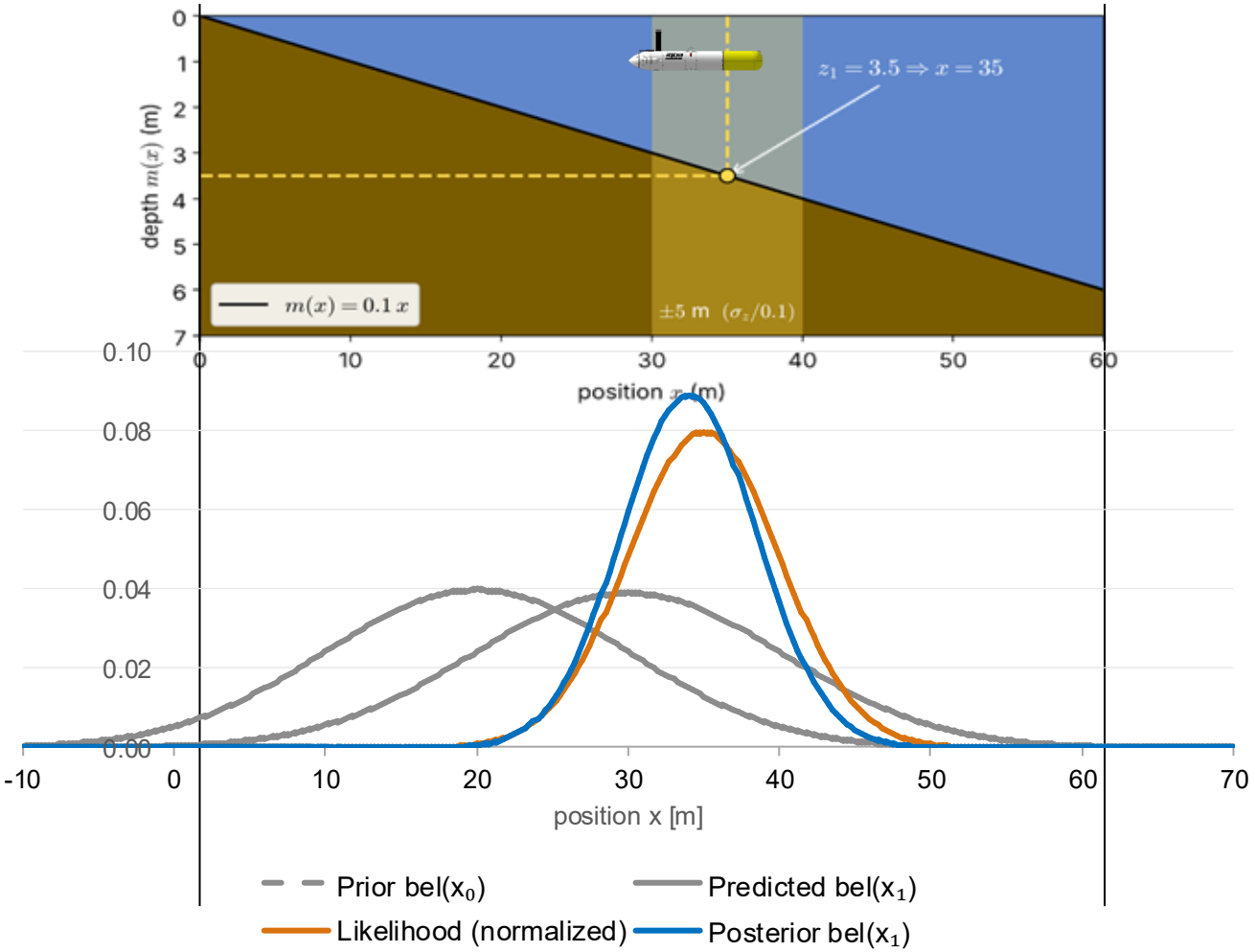
AUV Example: One Filter Iteration



Altitude: map m(x) = 0.1x $p(z_k \mid x_k) = \mathcal{N}(z_k;\, 0.1\,x_k,\, 0.5^2)$

Continuous Probability

AUV Example: One Filter Iteration

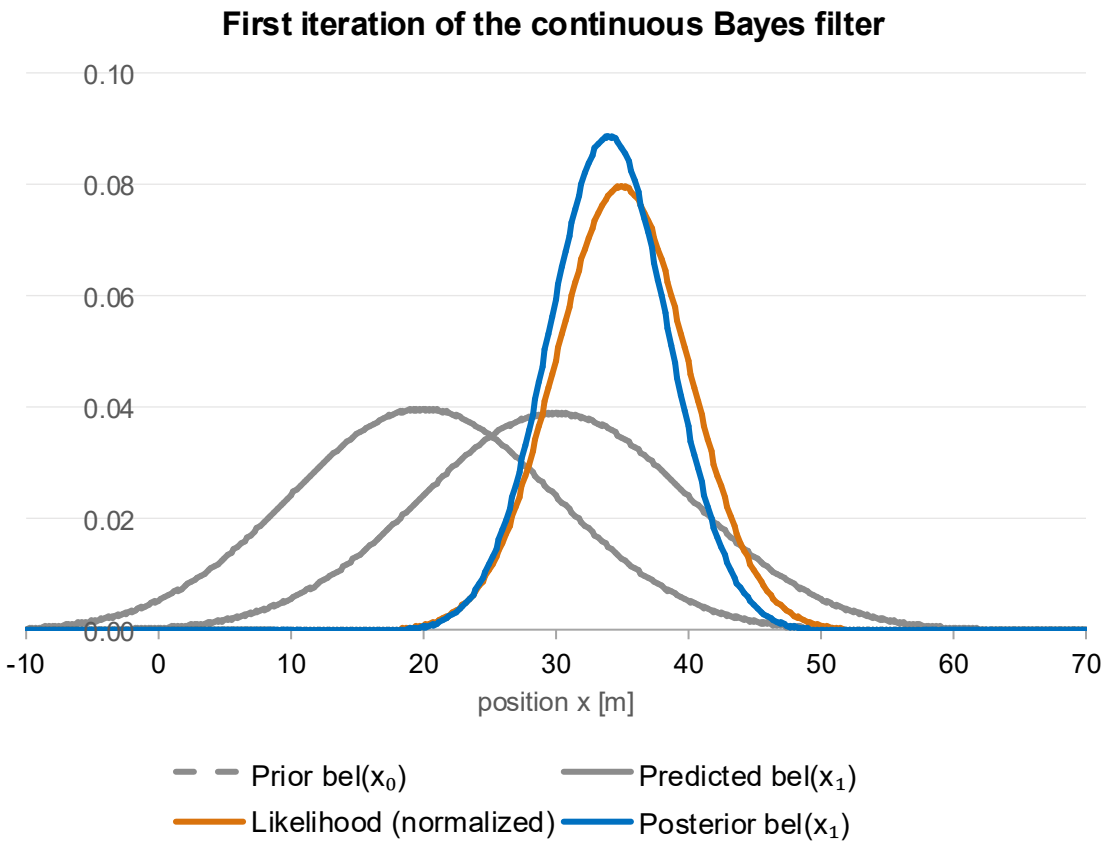


Update with z_1 = 3.5 (map depth at x = 35)

$$z_1 = 3.5 \;\Rightarrow\; p(z_1 \mid x) \propto \mathcal{N}(x;\, 35,\, 5^2) \quad \text{bel}(x_1) = \mathcal{N}(x;\, 34.0,\, 20.2)$$

AUV Example: One Filter Iteration

Prediction spreads the belief; the altitude reading sharpens it



0.040

Prior peak density [m^{-1}]

0.039

Predicted peak: lower and wider

4.5 m

Posterior σ, down from 10.2 m

Joint Density of a Trajectory

The joint density of a trajectory is a product of model terms

$$p(x_0, x_1, z_1) = p(x_0)\, p(x_1 \mid x_0, u_1)\, p(z_1 \mid x_1)$$

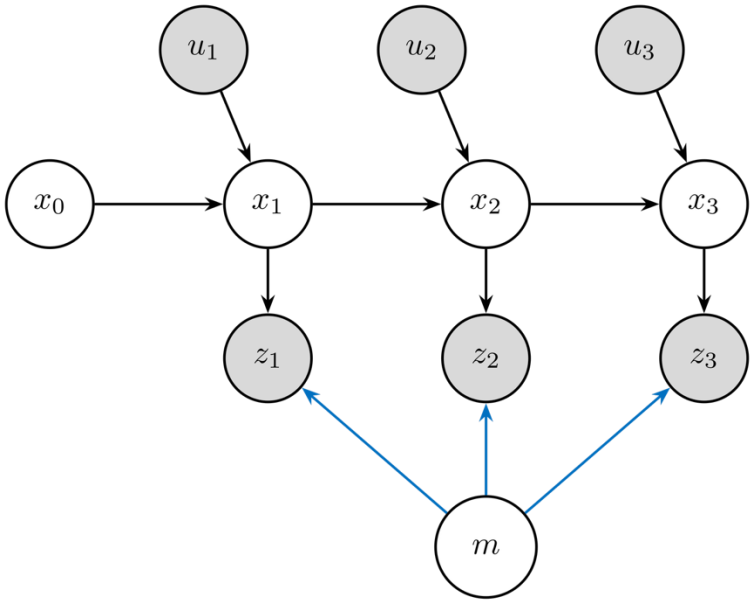
At $x_0 = 20$, $x_1 = 30$, $z_1 = 3.0$

$$0.0399 \cdot 0.1995 \cdot 0.7979 \approx 0.00635 \ \mathrm{m}^{-3}$$

- Each factor is a density with units m^{-1}, so the product has units m^{-3}.
- It is not a probability: that needs integration over a volume $dx_0\, dx_1\, dz_1$.

Taxonomy of Robot Localization and Mapping

Four problems, depending on what is known and what is estimated



Gray: observed (u, z). White: hidden (x, m).
The map m links every observation.

Localization

map known $p(x_k \mid z_{0:k}, u_{0:k}, m)$

Mapping

poses known $p(m \mid z_{0:k}, x_{0:k})$

Online SLAM (filtering)

current pose + map $p(x_k, m \mid z_{0:k}, u_{0:k})$

Full SLAM (smoothing)

whole trajectory + map $p(x_{0:k}, m \mid z_{0:k}, u_{0:k})$

Conclusions and Next Steps

The Bayes filter is exact on paper and approximate in practice

1. **Belief** — A pdf over a continuum, possibly multimodal on ambiguous terrain.
2. **Recursion** — Prediction with Chapman–Kolmogorov, correction with Bayes' rule.
3. **Problem** — No closed form for general pdfs and integrals.

NEXT: APPROXIMATE FILTERS

- **Non-parametric:** histogram (grid) filter, particle filter.
- **Parametric:** Gaussian filters, where μ and Σ describe the belief.